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LAPORAN TUGAS PRAKTIKUM 14

13.2 Persiapan

13.3 Praktikum - Implementasi ArrayList

class customer11.java

```
package praktikum14;

public class customer11 {
    public int id;
    public String name;

    public customer11(){

    }

    public customer11(int id,String name){
        this.id = id;
        this.name = name;
    }

    public String toString(){
        return " ID: " + this.id + " Nama: " + this.name;
    }
}
```

class book11.java

```
package praktikum14;

public class book11 {
    public String isbn;
    public String title;

    public book11(){

    }

    public book11(String isbn, String title){
```

```

        this.isbn = isbn;
        this.title = title;
    }

    public String toString(){
        return "ISBN: " + this.isbn + "Title: " + this.title;
    }
}

```

class demoArrayList11.java

```

package praktikum14;
import java.util.ArrayList;
public class demoArrayList11 {
    public static void main(String[] args) {
        ArrayList<customer11> customers = new ArrayList<>(2);

        customer11 customer1 = new customer11(1, "Zaskia");
        customer11 customer2 = new customer11(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new customer11(4, "Cica"));
        for (customer11 cust : customers){
            System.out.println(cust.toString());
        }
    }
}

```

5. Compile dan run kode program, di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?

Di akhir collection

ID: 1 Nama: Zaskia ID: 5 Nama: Budi ID: 4 Nama: Cica
--

7. Compile dan run kode program. Index pada ArrayList dimulai dari 0 atau 1?

```

customers.add(2, new customer11(100, "Rosa"));

```

jawab: index dimulai dari 0

ID: 1 Nama: Zaskia
ID: 5 Nama: Budi
ID: 100 Nama: Rosa
ID: 4 Nama: Cica

10. Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

jawab: Ya, ArrayList dapat diinstansiasi tanpa menentukan size di awal.

1
Budi
ID: 1 Nama: Zaskia
ID: 5 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica

11. Anda juga dapat menambahkan sekumpulan customer baru ke dalam ArrayList secara sekaligus. Misalnya terdapat ArrayList newCustomers. Tambahkan seluruh object customer sekaligus ke dalam customers.

class demoArrayList11.java

```
package praktikum14;
import java.util.ArrayList;
public class demoArrayList11 {
    public static void main(String[] args) {
        ArrayList<customer11> customers = new ArrayList<>();

        customer11 customer1 = new customer11(1, "Zaskia");
        customer11 customer2 = new customer11(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new customer11(4, "Cica"));

        customers.add(2, new customer11(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        customer11 customer = customers.get(1);
        System.out.println(customer.name);
        customer.name = "Budi Utomo";

        ArrayList<customer11> newCustomers = new ArrayList<>();
```

```

newCustomers.add(new customer11(201, "Della"));
newCustomers.add(new customer11(202, "Victor"));
newCustomers.add(new customer11(203, "Sarah"));

customers.addAll(newCustomers);

for (customer11 cust : customers){
    System.out.println(cust.toString());
}
}

```

hasil output

```

1
Budi
ID: 1 Nama: Zaskia
ID: 5 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
ID: 201 Nama: Della
ID: 202 Nama: Victor
ID: 203 Nama: Sarah

```

13.4 Praktikum - Implementasi Stack

class stackDemo11.java

```

package praktikum14;
import java.util.Stack;
public class stackDemo11 {
    public static void main(String[] args) {
        book11 book1 = new book11("1234", "Dasar Pemrograman");
        book11 book2 = new book11("7145", "Hafalah Shalat Delisa");
        book11 book3 = new book11("3562", "Muhammad Al-Fatih");

        Stack<book11> books = new Stack<>();
        books.push(book1);
        books.push(book2);
        books.push(book3);

        book11 temp = books.peek();

        if (temp != null){
            System.out.println(temp.toString());
        }
    }
}

```

```

    }

    book11 temp2 = books.pop();

    if (temp2 != null){
        System.out.println(temp2.toString());
    }
}
}

```

hasil output

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih

```

5. Mengapa perlu ada pengecekan (temp != null)?

jawab: Pengecekan temp != null diperlukan untuk mencegah program error ketika objek kosong, sehingga toString() hanya dijalankan jika variabel temp benar-benar memiliki data.

6. Lakukan looping untuk mencetak data buku pada stack

hasil output

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemrograman
ISBN: 7145 Title: Hafalah Shalat Delisa

```

8. Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

jawab: tambah code ini

```

int posisi = books.search(book2);
System.out.println("Buku ini berada di posisi " + posisi);

```

hasil output

```

ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 3562 Title: Muhammad Al-Fatih
ISBN: 1234 Title: Dasar Pemrograman
ISBN: 7145 Title: Hafalah Shalat Delisa
Buku ini berada di posisi 1

```

13.5 Praktikum - Implementasi TreeSet

class treeSetDemo11.java

```
package praktikum14;
import java.util.TreeSet;
public class treeSetDemo11 {
    public static void main(String[] args) {
        TreeSet<String> fruits = new TreeSet<>();

        fruits.add("Mangga");
        fruits.add("Apel");
        fruits.add("Jeruk");
        fruits.add("Jambu");

        for (String temp : fruits){
            System.out.println(temp);
        }
    }
}
```

4. Compile dan run program. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

jawab: TreeSet mengurutkan data String berdasarkan urutan alfabet, sehingga output menjadi

```
Apel
Jambu
Jeruk
Mangga
```

6. Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

jawab:

- first() :=mengambil elemen pertama (terkecil)
- last() = mengambil elemen terakhir (terbesar)
- remove() = menghapus elemen tertentu dari TreeSet.
- pollFirst() = mengambil dan menghapus elemen pertama
- pollLast() = mengambil dan menghapus elemen terakhir

class treeSetDemo11.java

```
System.out.println("First: " + fruits.first());
System.out.println("Last: " + fruits.last());
```

```

    fruits.remove("Jeruk");
    System.out.println("Setelah remove " + fruits);

    fruits.pollFirst();
    System.out.println("Setelah poll first " + fruits);

    fruits.pollLast();
    System.out.println("Setelah poll last " + fruits);

```

hasil output

```

Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah remove [Apel, Jambu, Mangga]
Setelah poll first [Jambu, Mangga]
Setelah poll last [Jambu]

```

13.6 Praktikum – Sorting

class demoArrayList11.java

```

package praktikum14;
import java.util.ArrayList;
import java.util.Collections;
public class demoArrayList11 {
    public static void main(String[] args) {
        ArrayList<customer11> customers = new ArrayList<>();

        customer11 customer1 = new customer11(1, "Zaskia");
        customer11 customer2 = new customer11(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);

        customers.add(new customer11(4, "Cica"));

        customers.add(2, new customer11(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        customer11 customer = customers.get(1);
    }
}

```

```

System.out.println(customer.name);
customer.name = "Budi Utomo";

ArrayList<customer11> newCustomers = new ArrayList<>();
newCustomers.add(new customer11(201, "Della"));
newCustomers.add(new customer11(202, "Victor"));
newCustomers.add(new customer11(203, "Sarah"));

customers.addAll(newCustomers);

for (customer11 cust : customers){
    System.out.println(cust.toString());
}

ArrayList<String> daftarSiswa = new ArrayList<>();
daftarSiswa.add("Zainab");
daftarSiswa.add("Andi");
daftarSiswa.add("Rara");
Collections.sort(daftarSiswa);

System.out.println(daftarSiswa);

customers.sort((c1, c2) -> c1.name.compareTo(c2.name));
}
}

```

hasil output

```

1
Budi
ID: 1 Nama: Zaskia
ID: 5 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
ID: 201 Nama: Della
ID: 202 Nama: Victor
ID: 203 Nama: Sarah
[Andi, Rara, Zainab]

```